# Alvin Lionel Thumi Thooko

Software Engineering student

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https://lionel-portfolio-one.vercel.app/

GitHub:

https://github.com/AlvinLionel


## Summary


Software Engineering student with hands-on experience building full-stack applications and software systems using React, TypeScript, Node.js, Express, C, PHP and SQL. Experienced in developing REST APIs, integrating third-party services, working with relational and NoSQL databases, and building security-focused applications.


## Tech stack

**Languages:** C, JavaScript, TypeScript, SQL

**Frontend:** HTML, CSS, Tailwind, React, Next.js

**Backend:** Node.js, Express

**Databases:** MySQL, PostgreSQL, MongoDB

**APIs & Integration:** REST APIs, API Integration, Safaricom Payment Daraja API

**Security & Cryptography:** Web Crypto, AES, RSA, ECDH/X25519

**Tools:** Git, GitHub, Vite, npm, prisma

**Engineering:** Requirements Engineering, Database management, Technical Documentation, Version Control

## Projects

GitHub: https://github.com/AlvinLionel/My_Projects

**LockBoxX-** Browser-based security system | *React, TypeScript, Web Crypto API*

Built a client-side security platform for encrypting and decrypting multiple resource types and with native-password support for some resources.

Implements symmetric algorithms, public keys, key exchange algorithms and native-password protection techniques to secure resources.

https://lockboxx.vercel.app/

**Full-Stack E-commerce Platform | Ongoing** | Node.js, Express, React, TypeScript, MySQL

Developing a full-stack e-commerce platform for discovering home products, nearby stores and home improvement services.

Implemented user authentication, product catalog, shopping cart functionality and REST API endpoints with an integrated interactive map functionality.

**Checkout System** | JavaScript, Node.js, MySQL, Bootstrap, Safaricom Daraja API

Built a store checkout page with product and transaction data managed through MySQL

Integrated Safaricom Daraja to support mobile payment and processing.

**Static Server** | Node.js

Built a custom static file server using Node.js for serving files over http.

Implemented event and error logging to monitor server activity and diagnose failures.

**Weather Application** | HTML, CSS, JavaScript, REST API

Developed a dynamic weather interface that extracts real-time data from an external API.

Implemented responsive weather displays, cards and charts based on API responses and user requests

**Others**

https://sbgcuk.pages.dev/

## Education

The Co-operative University of Kenya

**Bachelor of Science in Software Engineering**

*Expected Graduation: 2028.*